

IBM HR Analytics — Portfolio Cheat Sheet

Quick reference for the entire end-to-end attrition analysis project

SECTION 1 — PROJECT OVERVIEW

What: End-to-end analysis of employee attrition using IBM HR Analytics dataset (1,470 employees, 35 features). **For whom:** CEO and CHRO. **Goal:** Reduce attrition from 16.1% to below 10% with actionable, cost-justified interventions.

SECTION 2 — BUSINESS CONTEXT & OBJECTIVE

Problem: Attrition rate 16.1% — above industry average (13-17%) and well above best-practice target ($\leq 10\%$). 1 in 6 employees leaving. **Cost:** Replacing one employee costs 50-200% of annual salary (SHRM). **Objective:** Identify drivers, predict risk, deliver 3-4 targeted interventions with cost-benefit analysis.

SECTION 3 — BUSINESS QUESTIONS

Q	Question	Answered In
1	Why are employees leaving, and which groups are most at risk?	EDA + Model
2	Can we identify employees likely to leave before they resign?	Modeling + Risk Scoring
3	What are the most cost-effective retention actions?	Intervention + Impact
4	How do we validate that interventions actually work?	A/B Test + Monitoring

SECTION 4 — DATA LOADING & UNDERSTANDING

Dataset: 1,470 rows x 35 columns. Cross-sectional snapshot (not time-series). Source: IBM HR Analytics (Kaggle). 26 numerical + 9 categorical features. **Target:** Attrition (Yes/No $\rightarrow$ 1/0). **Imbalance:** 84% Stay vs 16% Leave.

SECTION 5 — DATA CLEANING & PREPROCESSING

Dropped: EmployeeCount, Over18, StandardHours (constant values), EmployeeNumber (identifier). **No issues:** No missing values, no duplicates, no typos in categoricals. **Outliers:** Present but retained — contextually valid (salary range reflects job level hierarchy). **Final shape:** 1,470 rows x 31 columns.

SECTION 6 — EXPLORATORY DATA ANALYSIS (EDA)

Approach: Domain-driven variable selection (not correlation-driven). Focused on 7-8 high-impact variables across 3 HR dimensions: workload, career stage, compensation. Validated with Chi-Square tests (all $p < 0.05$).

Key EDA Findings:

Driver	Key Finding	Attrition Rate
Overtime	Strongest single driver — 3x higher attrition	30.5% (vs 10.4% no OT)
Early Career	0-2 yrs experience = highest risk window	43.9%
First Year Tenure	First year is critical retention gap	34.9%
Low Salary (Q1)	Bottom quartile = 3x higher than top	29.3%
OT + Low Salary	DEPARTURE TRIGGER — more than half leave	58.5%
No Stock Options	2.5x higher than Level 1+	24.4% (vs 9.9%)

SECTION 7 — FEATURE ENGINEERING

3 types of features: Encoding (OverTimeFlag: Yes/No → 1/0), Binning (ExperienceGroup, TenureGroup, SalaryGroup — for EDA only, raw values used in model), New features (OverTime_LowSalary interaction, EarlyCareer composite flag). **Key insight:** Binning helps visualization but model benefits from raw continuous values.

SECTION 8 — MODELING

Why Logistic Regression: Interpretability for CEO/CHRO. Coefficients → odds ratios → business language. **Features:** 9 features (3 engineered + 6 raw). StandardScaler applied for convergence.

Model Comparison:

Metric	LR Baseline	LR Balanced	RF Balanced
Accuracy	85.4%	71.8%	77.6%
Recall (Attrition)	21.3%	61.7%	42.6%
ROC-AUC	0.740	0.742	0.749
PR-AUC	0.467	0.476	0.403

Decision: Balanced Logistic Regression selected — best recall (62%), best PR-AUC (0.476), full interpretability. RF benchmark confirms performance ceiling ~0.74 AUC — adding complexity doesn't help. **Business justification:** Missing a departure (cost: 50-200% salary) >> false alarm (cost: a conversation).

SECTION 9 — MODEL INTERPRETATION

Odds Ratios (what to tell the CEO):

Feature	Odds Ratio	Business Translation
OverTimeFlag	1.57	OT employees are 57% more likely to leave
EarlyCareer	1.56	First 1-2 years employees 56% more likely to leave
OverTime_LowSalary	1.31	OT + low salary adds 31% compounding risk
StockOptionLevel	0.68	Each stock level reduces risk by 32%
MonthlyIncome	0.78	Higher income reduces risk by 22% per std dev

LogReg vs RF perspective: LogReg top 3 = OverTimeFlag, EarlyCareer, OT_LowSalary (directional effect). RF top 3 = MonthlyIncome, Age, TotalWorkingYears (overall predictive power). Not contradictory — complementary. **EDA consistency:** All 5 key drivers confirmed by model — full alignment.

SECTION 10 — RISK SCORING

Method: Score all employees 0-100% probability, then categorize:

Risk Category	Threshold	Count	% Workforce	Actual Att. Rate
High Risk	≥ 0.7	181	12.3%	51.9%
Medium Risk	0.4 - 0.7	487	33.1%	18.1%
Low Risk	< 0.4	802	54.6%	6.9%

Key proof: High Risk group (12.3% of workforce) has 51.9% actual attrition — 7x higher than Low Risk (6.9%). High + Medium combined captures 76.8% of all attrition cases. **Profile:** 77% overtime, avg income \$2,994 (half of company avg), 61% early career, 67% no stock options.

SECTION 11 — DECISION FRAMEWORK

Purpose: Translate risk scores into operational HR actions. Model is a prioritization tool, not a decision-maker. **Key additions:** Priority ranking within High Risk (P1-P4 based on overlapping risk signals), 6-step operational workflow (score → list → assign → interview → recommend → track), cost of inaction for High Risk group: \$1.7M-\$3.4M in replacement costs.

SECTION 12 — BUSINESS INTERVENTION

Goal: Prevent 90+ departures to reach <10% attrition. 4 targeted interventions ranked by impact/dollar:

P#	Intervention	Target	Prevented	Cost	Savings
1	Overtime Reduction	416 emp (140 HR)	~83	Low (process)	\$3.2M-\$6.5M
2	Stock Option Grant	631 emp (122 HR)	~91	Med (equity)	Significant
3	Early-Career Retention	242 emp (110 HR)	~51	Low (~\$50K)	~\$1.0M
4	Compensation Review	138 emp (HR+Low)	Support	\$470K-\$700K	In #1-3
	COMBINED		90+	\$520K-\$750K	\$4M-\$7M+

Intervention Logic:

P1 Overtime: Strongest driver (OR=1.57). Process change, not cash. Audit teams, cap sustained OT, redistribute workload. Success: OT attrition 30.5% → <15%.

P2 Stock Options: Strongest protective factor (OR=0.68). Grant Level 1 to high-risk employees. Vesting tied to tenure milestones. Equity-based = retention anchor. Success: no-stock attrition 24.4% → <12%.

P3 Early-Career: Redesign first 90 days. Mentor/buddy program. Stay interviews at M1, M3, M6. Career path visibility early. Success: first-year attrition 34.9% → <20%.

P4 Compensation: Targeted 10-15% raise for 138 Low Salary + High Risk (not company-wide). Combine with stock grant for max effect. Partial closure \$470K-\$700K vs full \$2.7M. Success: low-salary attrition 29.3% → <15%.

SECTION 13 — EXPERIMENT DESIGN PROPOSAL

Purpose: Validate interventions before full rollout. These are proposals, not executed experiments (data is historical). **Design:** Each intervention has Treatment (A) vs Control (B) group, primary metric (attrition rate), sample size (~50-70 per group), and pilot duration (6-12 months). **Decision rule:** Significant reduction in attrition ($p < 0.05$) → proceed with rollout.

SECTION 14 — MONITORING & ITERATION

6 KPIs: Overall attrition rate, attrition by risk category, OT exposure rate, first-year attrition, high-risk count, intervention completion rate. Each with baseline, target, frequency, and owner. **Model retrain triggers:** Recall <50%, new attrition patterns, org changes, or every 12 months minimum. **Intervention monitoring:** Per-intervention success/failure signals with review points (6-12 months). **Principle:** Closed feedback loop — score → intervene → track → retrain → repeat.

SECTION 15 — CONCLUSION

3 core drivers: Overtime (OR=1.57), Early Career (OR=1.56), Compensation (OR=0.68-0.78). **Bottom line:** \$520K-\$750K investment → \$4M-\$7M+ savings. Attrition 16.1% → <10% achievable. **Model:** Prioritization tool, not crystal ball. Value = complete system (model → score → decide → intervene → monitor).

KEY NUMBERS TO REMEMBER

Metric	Value
Current Attrition Rate	16.1% (237 of 1,470 employees)
Target Attrition Rate	< 10% (need to prevent 90+ departures)
Overtime Attrition Rate	30.5% vs 10.4% (no OT) — 3x difference
Deadliest Combo	OT + Low Salary = 58.5% attrition (106 employees)
Model Recall	62% — catches 3 in 5 at-risk employees
High Risk Group	181 employees (12.3%) with 51.9% actual attrition
Coverage	High + Medium Risk captures 76.8% of all attrition
Strongest Risk Factor	OverTime — OR 1.57 (57% more likely to leave)
Strongest Protective Factor	StockOptionLevel — OR 0.68 (32% risk reduction per level)
Investment Needed	~\$520K-\$750K + equity allocation
Replacement Cost Saved	\$4M-\$7M+ in avoided replacement costs
ROI	Strongly positive — interventions pay for themselves 5-10x

SECTION STRUCTURE (15 SECTIONS)

Phase	Sections
Problem Framing	1. Project Overview → 2. Business Context → 3. Business Questions
Data Work	4. Data Loading → 5. Data Cleaning → 6. EDA (with Chi-Square)
Modeling	7. Feature Engineering → 8. Modeling (LR + RF) → 9. Interpretation
Operationalization	10. Risk Scoring → 11. Decision Framework → 12. Interventions
Validation & Closure	13. Experiment Design → 14. Monitoring → 15. Conclusion